Mass-Radius Relation in Exoplanets

Nisa Nur Güneş - Izmir Atatürk High School

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1 Abstract

Exoplanet research is one of the most current and important areas of research in modern astronomy and astrophysics. The study of these planets outside our solar system not only demonstrates how diverse and vast the universe is, but also helps us to understand topics such as planetary formation and evolution, planetary types and structures, and potential habitability conditions. This project involved a study of exoplanets that have been confirmed in the literature and for which mass-radius data are available. The data analyzed within the scope of our project were examined using Microsoft Excel to determine the relationship between mass and radius. The graphs drawn based on the data obtained were also analyzed using this program. Our analyzes revealed a varying mass-radius relationship for different types of exoplanets, resulting in three different equations for small, medium, and large-mass exoplanets. These equations are a crucial tool that allows us to estimate the other quantity when either the mass or the radius is available, thereby providing insight into the type and structure of the exoplanet. In this study, functional equations were obtained separately for each class of planets characterized by breakpoints determined using a simpler and more applicable method.

2 Introduction

All the planets in our Solar System orbit around the Sun in specific orbits. Like our Sun, other stars may have their planets orbiting around them. In this way, planets around stars other than the Sun are called exoplanets.

The first planet outside the Solar System was discovered in 1992 (Wolszczan and Frail, 1992). The discovery of this exoplanet was made using the vertical velocity measurement method. This method is based on the measurement of shifts in the spectral lines of the planet's orbital motion around its host star as it moves closer and farther away from the observer. Exoplanets are also being discovered in other ways. The most important of these is the transit method, in which exoplanets are observed as they pass in front of their host stars and the decrease in the star's light is observed. Exoplanets are very difficult to observe directly. This is because they orbit a star whose size and brightness make them difficult to observe. However, as an exoplanet passes in front of its star, the amount of light that reaches us from the star is slightly reduced. By measuring the amount of light reaching us from the star, we can determine whether there is a planet orbiting that star. This method, which is widely used to explore exoplanets, is called the transit method. Vertical velocity and transit methods are the two main methods used in the discovery of exoplanets and obtaining their parameters. Exoplanet discoveries can be made using methods other than these two, but these are the most widely used. The discovery and study of exoplanets are very important in terms of planet formation and evolution, as well as planet characterization.

While the first exoplanets discovered were planets with relatively large masses and radii, many smaller planets have been discovered in recent years due to advances in technology and observational instruments. In particular, the introduction of space telescopes has been an important stepping stone for the discovery of different types of exoplanets. The most important instruments used in exoplanet discoveries are the Kepler space mission (Borucki et al., 2010), CoRoT (Baglin et al., 2006), the Transiting Planet Survey Satellite (TESS), the James Webb Space Telescope and many ground-based observatories. Thanks to the ever-advancing technology and the space telescopes that are the products of this technology, more than 7300 exoplanets have been discovered today (www.exoplanet.eu). This number is increasing every day.

Traditionally, planets have been divided into two main groups. The first group includes large gas-dominated planets, while the second consists of small terrestrial planets (e.g. Weidenschilling, 1977). This distinction is inspired by the solar system, where large planets 3 are made of volatile matter (e.g. Jupiter), while terrestrial planets are small and made of durable matter. It is important which parameter (mass or radius) is used to distinguish between small and large planets. If a classification is to be made according to mass, it is generally made from $\tilde{3}0$ $M_{M_\oplus}$ (the symbol $M_\oplus$ is used to indicate the Earth. Here $M_{M_\oplus}$ stands for the mass of the Earth.) Planets with a mass smaller than $M_{M_\oplus}$ are considered minor planets (Mayor et al., 2011; Howard et al., 2010), while planets with a radius smaller than 4 $R_{M_\oplus}$ are included in this group (Marcy et al., 2014; Weiss and Marcy, 2014).

As the study of exoplanets has increased, various subclasses have emerged under these two main planetary classifications. Gas giants are massive planets, typically composed of gas (primarily helium or hydrogen), with small rocky cores. Exoplanets of this type are similar to Jupiter and Saturn in both size and lack of a solid surface. Most planets with masses exceeding 10 Earth masses are called gas giants. When gas giants are closer to stars, they are called "Hot Jupiters/Saturns" and their temperatures can reach thousands of degrees Celsius. The group of giant planets that are mostly composed of heavier elements, rather than Hydrogen and Helium (H/He), are called Neptune-class planets. These planets still have an envelope of H/He, but this lighter elemental envelope does not make up most of the planetary mass. Examples of such planets in our Solar System are Uranus and Neptune. There is also another group of exoplanets within the type, called mini-Neptune planets, which are larger than Earth but smaller than Neptune. Water, ammonia, and methane are usually found frozen as ice in the outer parts of the solar system. Because Uranus and Neptune are made of these three chemicals, they are also called "Ice Giants".

Planets called super-Earths are larger than the Earth. These planets can be twice the size of the Earth in terms of size (radius) and ten times the mass of the Earth. But these planets are smaller than Neptune. They are lighter than ice giants and can be composed of rock or gas or a combination of these

(potentially rocky surfaces). They are called super Earths not because they look like Earth, but because of their large size. Planets of this type are alien to us because they have no analogs in our Solar System. Exoplanets at the maximum size of super-Earths are called sub-Neptunes or mini-Neptunes, as mentioned above.

Earth-like or terrestrial planets are rocky planets such as Mars, Venus, and Mercury, composed predominantly of elements such as Silicon, Magnesium, Iron, Oxygen, and Carbon 4 (Sotin et al., 2007) and have a largely liquid or solid surface. These planets may also have an atmosphere made of gases. More research is needed to see if these exoplanets show signs of habitability with their atmospheres, oceans, etc. It is these terrestrial planets that hold the most hope for life. In the exoplanet literature, apart from these types, there are also planets called "Ocean Planets". Water is a crucial compound for the formation and development of life. There is evidence that some of the moons of the giant planets in our solar system (such as Callisto, Ganymede, Europa, Titan, and Enceladus) harbor subsurface liquid water. These findings have led to research on the possibility of ocean planets. It is not yet known whether oceanic planets exist, but there are efforts to determine their theoretically possible properties (Léger et al., 2004).

By examining the relationships between the parameters of exoplanets, crucial conclusions can be drawn about their formation, type, and evolution. For example, the relationship between mass and radius, which we will examine in this project, provides information about the composition and structure of exoplanets. Determining the mass-radius (M - R) relationship and how this relationship varies for different types of planets not only provides us with important information about the structure and type of planets but is also important for understanding their formation scenarios. Relating a measured radius to a planet's mass can also play a role in vertical velocity tracking by estimating the expected vertical velocity semi-amplitude of a transiting exoplanet.

Some studies examining the $M - R$ (Mass-Radius) relationship have suggested that there is a transition between small planets (Neptune-like) and large planets (Jupiter-like) in the $M - R$ relationship. Weiss et al. (2013), based on a visual examination of the $M - R$ relationship, determined that this transition point corresponds to $150\,M_\oplus$ and, as a result of their analyses, found the $R_p \propto M^{0.54}$ relationship for planets smaller than $150\,M_\oplus$ and the $R_p \propto M^{-0.039}$ for planets above this mass value. Hatzes and Rauer (2015) conducted a similar study for the mass-density relationship and determined the breakpoint to be $\sim 0.95\,M_J$ (or $M_\oplus$ as per context). Chen and Kipping (2017), in their study on the $M - R$ relationship, proposed $\sim 130 \pm 22\,M_\oplus$ for the transition between Neptune- and Jupiter-type planets and obtained the relationships $R_p \propto M^{0.59}$ and $R_p \propto M^{-0.04}$ for small and large planets, respectively. Although the masses proposed for the transition between small and large planets differ in these various studies, these given masses are larger than the traditionally accepted small

planet boundary ($\sim$ 20–30 $M_\oplus$). Bashi et al. (2017) updated this mass boundary to $124 \pm 7\,M_\oplus$ and proposed the relationship $R_p \propto M^{0.55\pm0.02}$ for small planets and $R_p \propto M^{0.01\pm0.02}$ for large planets. Finally, Müller et al. (2024) re-examined the $M - R$ relationship and transit points using Planets data. Their work yielded the following results: $R \propto M^{0.27}$ for small planets up to $\sim 4.4\,M_\oplus$; $R \propto M^{0.67}$ for planets with masses between $\sim$ 4.4 and $127\,M_\oplus$, classified as medium-mass planets; and finally, $R \propto M^{0.06}$ for planets with masses greater than $127\,M_\oplus$. Most of the studies used a power law model to examine the mass-radius relationship.

Exoplanets have been discovered and studied for over 30 years. As a result of these studies, numerous investigations have been and continue to be conducted, primarily on the mass-radius relationship and the characterisation of exoplanets. Determining an exoplanet's mass-radius relationship can provide us with information about parameters such as the planet's density and surface gravity. This information allows us to obtain information about the type of planet (rocky, gaseous, etc.). The M-R relationship also helps us make predictions about the internal structure and chemical composition of the planet being studied. Information about the formation and evolution of the planet and the dynamics of the evolutionary process can be obtained. The mass-radius relationship plays a critical role in understanding whether a planet is habitable. In this project, the mass-radius relationships of exoplanets were analyzed using an easier, faster, and more applicable method. Characterisations and functional equations were derived from the identified breakpoints, contributing to the exoplanet literature.

3 Methodology

The primary method employed in our project is the 'Correlational Research' technique. The exoplanet mass and radius data required to analyze observational M-R relationships within this project were obtained from the database available at https://exoplanet.eu/catalog/. The data found in the exoplanet database at the relevant web address was first downloaded to a computer and opened in Microsoft Excel, an electronic spreadsheet programme. This programme allows for operations such as data organisation, calculation, graphing, and representation of data with certain functions when working with such numerical data. It even allows for some basic analysis and solutions.

The exoplanet database provides numerical data on the basic physical parameters (such as mass, radius, temperature) and orbital parameters (such as semi-major axis length, orbital eccentricity, orbital period, and inclination) of discovered and confirmed exoplanets and their host stars. Since the mass and radius data in the database are given in Jupiter mass and radius units, the units were first converted to Earth mass and radius units using the calculation features of Microsoft Excel. To do this, the mass of all exoplanets was simply multiplied by 317.8, which is the ratio of Jupiter's mass to Earth's mass

(1.898×10^{27} / 5.972×10^{24} kg), and their radii were multiplied by 11.2, which is the ratio of Jupiter's radius to Earth's radius (69,911 / 6,378 km). Following this unit conversion, the data filtering feature of Microsoft Excel was used to identify the exoplanets to be used in the M-R (mass-radius) relationship analysis. Out of a total of 7,331 exoplanets, only those with available mass and radius data were selected, and 1,993 exoplanets were analyzed. Using this remaining data, an M-R graph was created using the graphing feature of Microsoft Excel, thus enabling the visual examination of the mass and radius distribution. The resulting graph is shown in Figure 1. The horizontal and vertical axes are plotted on a logarithmic scale.

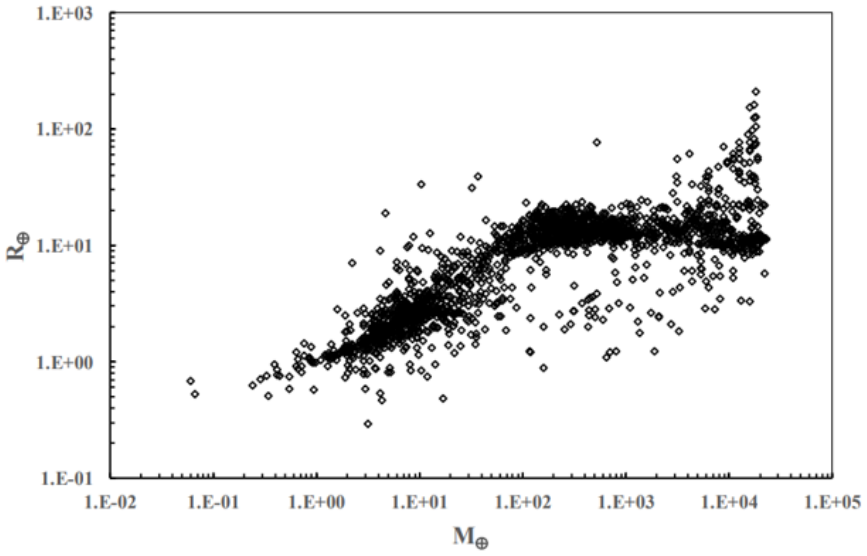


Figure 1: M – R graph created for exoplanets with available mass and radius data. In the figure, the horizontal axis represents exoplanet masses in Earth mass units, while the vertical axis represents exoplanet radii in Earth radius units.

As can be seen in Figure 1, exoplanets are distributed along a certain line on the M-R graph. However, points showing significant deviations from the general trend are noteworthy. In order to perform a more accurate analysis, these points with very large deviations were first removed from the data. After this process, we were left with mass-radius data for a total of 1751 planets. The M-R graph created with this remaining data is shown in Figure 2. Data for the planets of the Solar System has also been added to this graph. In this way, since we can also see the positions of the Solar System planets on the exoplanet M-R graph, we can compare these planets, about which we have detailed information

regarding their structural properties, with exoplanets.

As emphasised in the literature, the M-R graph given in Figure 2 shows a fragmented distribution. While some studies analyse it as two parts, small and large planets, recent studies (e.g. Müller 2024) have observed that it exhibits a three-part distribution (small, medium and large mass exoplanets) and analysed it with three-part function representations. Following a similar approach, we decided to perform the analysis assuming a three-part distribution with two breakpoints. For the analysis, we first attempted to determine the breakpoints in the distribution. Examination of the graph revealed that the first breakpoint was approximately 2.5 $M_\oplus$ and the second breakpoint was approximately 90 $M_\oplus$. Exoplanets with masses below 2.5 $M_\oplus$ were designated 'Small Mass Exoplanets', those with masses between 2.5 and 90 $M_\oplus$ are classified as 'Medium Mass Exoplanets', and finally, exoplanets with masses above 90 $M_\oplus$ are classified as 'Large Mass Exoplanets'.

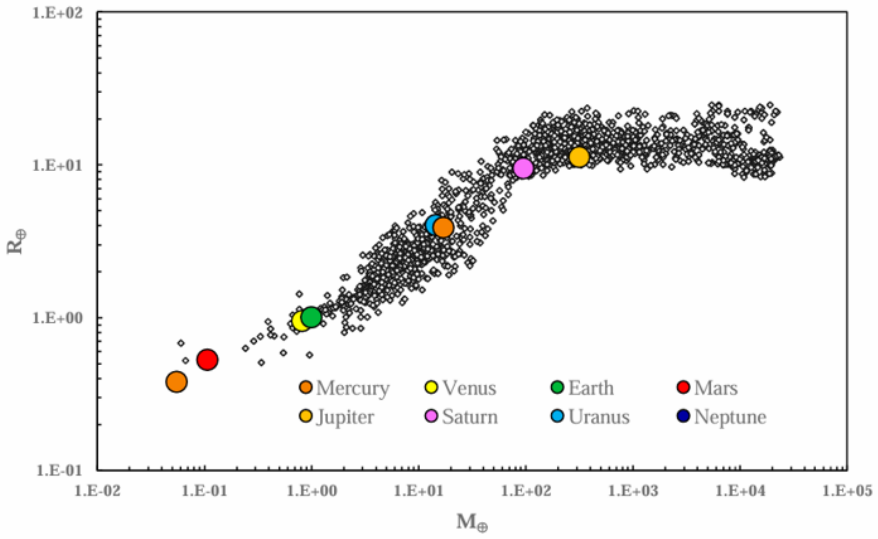


Figure 2: M–R graph created after removing the points with large scatter in Figure 1. The planets of the solar system have also been added to the graph with coloured and larger symbols.

In the next step of our analysis, we attempted to determine the M–R relationship for three separate regions, taking into account the breakpoints identified above (2.5 $M_\oplus$ and 90 $M_\oplus$). Three separate graphs were created for these regions using Microsoft Excel. We attempted to determine the function that best represented the data for each region. To do this, we used the Microsoft Excel program's feature for adding trend lines to data. This feature allows different

types of functions, such as exponential, linear, and logarithmic, to be added to the data. In existing analyses in the exoplanet literature, the M-R relationship has been determined by the power (exponent) law. This law assumes that one of the two quantities is related to the other by a certain exponent. The equation of the function represented by the trend line, added with the Microsoft Excel program, can also be printed. In the graphs created, an exponential trend line was added to the data for each region, and the equations of the functions represented by these lines were printed on the graph (Figure 3, Figure 4, and Figure 5). Thanks to these graphs, different mass ranges can be viewed in detail, both with exponential trend lines and with equations that give the relationship between mass and radius.

Figure 3 shows the M–R graph for low-mass exoplanets. This graph was created using 87 exoplanets with masses less than 2.5 $M_{\oplus}$, and an exponential trend equation obtained as R= $1.0114M^{0.2716}$ has also been added to the graph.

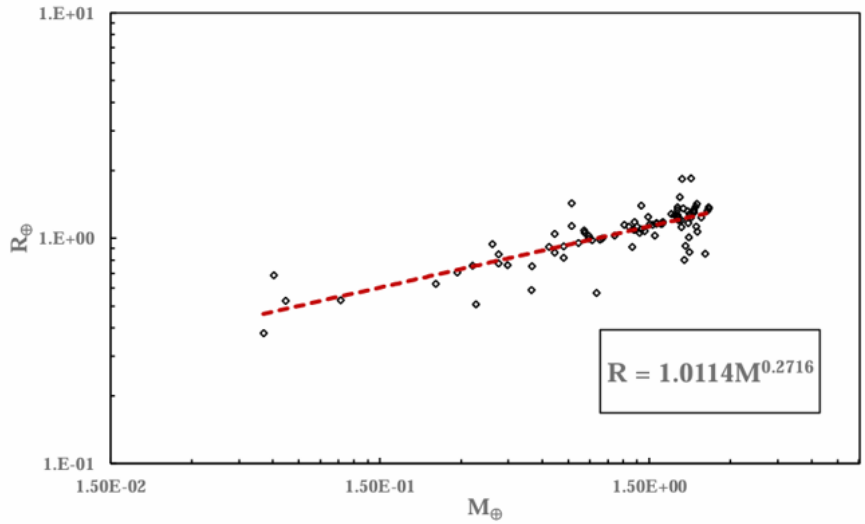


Figure 3: The exponential trend line for 87 exoplanets with a small mass (M < 2.5 $M_{\oplus}$) was generated by taking into account the breakpoints in Figure 2 and the M-R graph containing the equation of the function represented by this line.

Figure 4 shows the M–R diagram for medium-mass exoplanets. This graph was created using 665 exoplanets with masses ranging from 2.5 $M_{\oplus}$ to 90 $M_{\oplus}$, and the exponential trend equation obtained as R = $0.772M^{0.5659}$ has also been added to the graph.

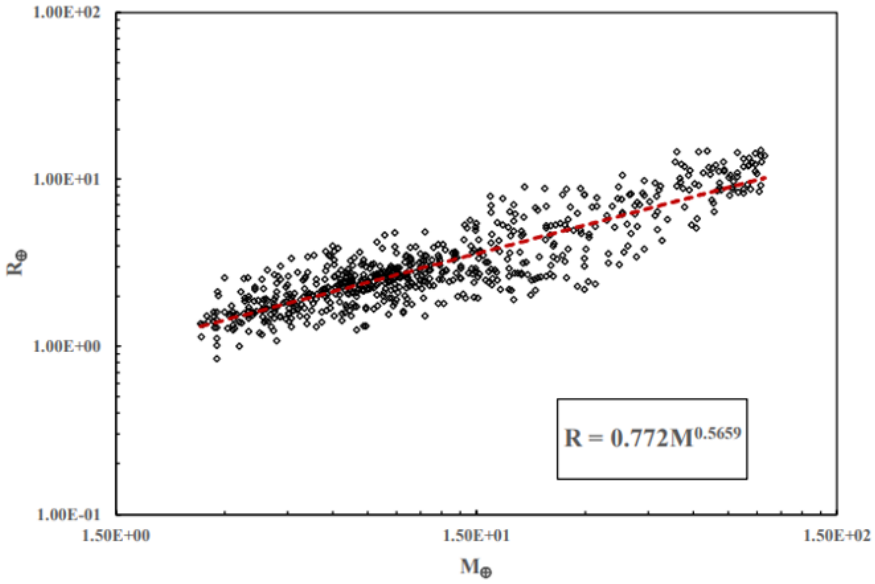


Figure 4: The exponential trend line of the intermediate mass 665 exoplanets ($2.5\ M_\oplus < M < 90\ M_\oplus$) was created by considering the breakpoints in Figure 2 and the M-R graph containing the equation of the function represented by this line.

Figure 5 shows the M–R graph for massive exoplanets. This graph was created using 1,007 exoplanets with masses greater than $90, M_\oplus$, and the exponential trend equation obtained as $R = 15.043, M^{-0.016}$ has also been added to the graph.

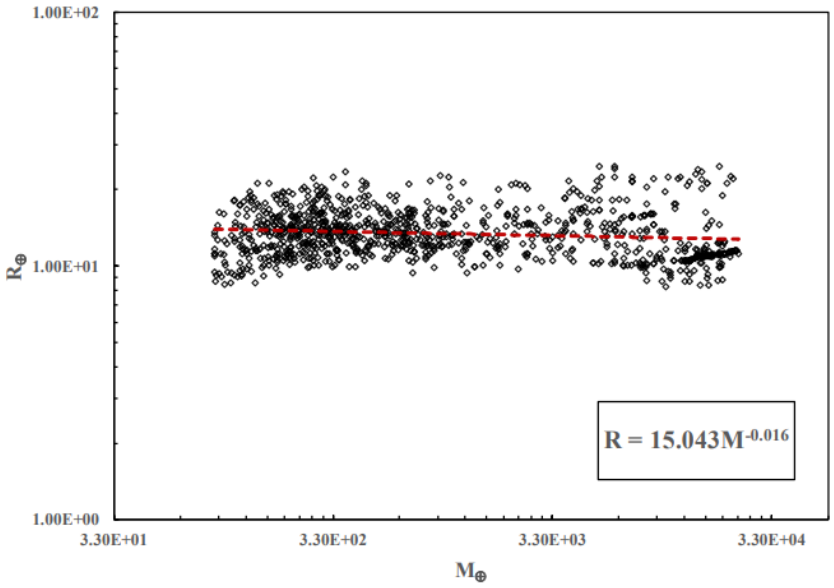


Figure 5: The M–R diagram showing the exponential trend line for 1007 high-mass exoplanets satisfying $90\,M_\oplus < M$, selected based on the breakpoints identified in Figure 2, along with the equation of the function representing this trend line.

The results obtained in the analysis are presented on a single graph in Figure 6. Small-mass planets are shown with black symbols, medium-mass planets with blue symbols, and large-mass planets with yellow symbols. The breakpoints of the M–R distribution ($2.5, M_\oplus$ and $90, M_\oplus$) are also shown on the graph with vertical dashed lines.

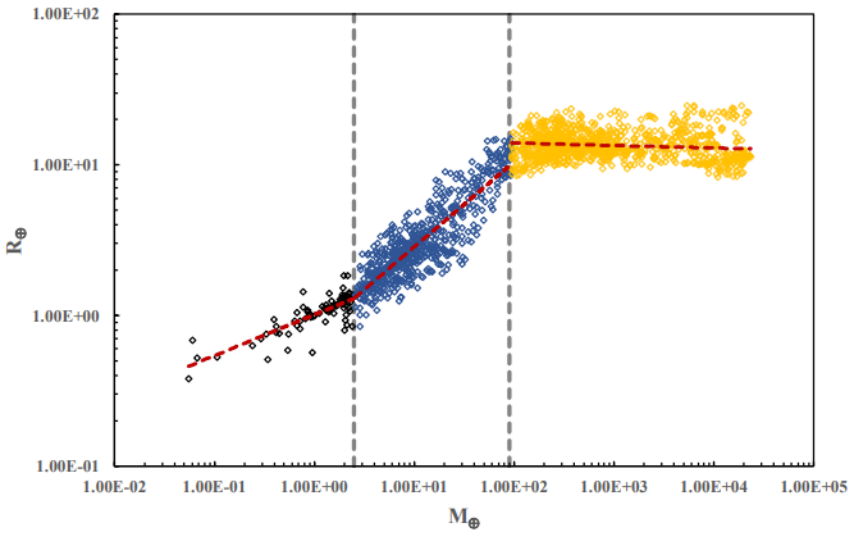


Figure 6: An M–R plot created by combining the graphs of low, medium, and high-mass exoplanets shown in Figures 3, 4, and 5.

4 Findings

In this study, we investigated the relationship between the masses and radii of exoplanets. As of the date on which the data were retrieved for this project, a total of 7,331 confirmed exoplanets had been identified, with mass and radius measurements available for 1,993 of them. An M–R distribution was constructed, and data points exhibiting substantial deviations from the overall distribution were excluded, leaving 1,751 data points for subsequent analysis. Two breakpoints were identified in the M–R distribution at $2.5\,M_{\oplus}$ and $90\,M_{\oplus}$, and the distribution was analyzed separately across three distinct mass ranges. For low-mass rocky planets, including Mercury, Venus, Earth, and Mars, with masses below $2.5\,M_{\oplus}$, the relationship $R = 1.00114\,M^{0.2716}$ was obtained. For intermediate-mass planets, including gas and ice giants such as Uranus and Neptune, as well as rocky planets classified as super-Earths, with masses between $2.5\,M_{\oplus}$ and $90\,M_{\oplus}$, the relationship $R = 0.772\,M^{0.5659}$ was derived. For more massive gas and ice giant exoplanets, such as Saturn and Jupiter, with masses greater than $90\,M_{\oplus}$, the relationship $R = 15.043\,M^{-0.016}$ was obtained. The derived equations, together with the mass–radius (M–R) distribution analyzed in this study, are presented in Figure 7.

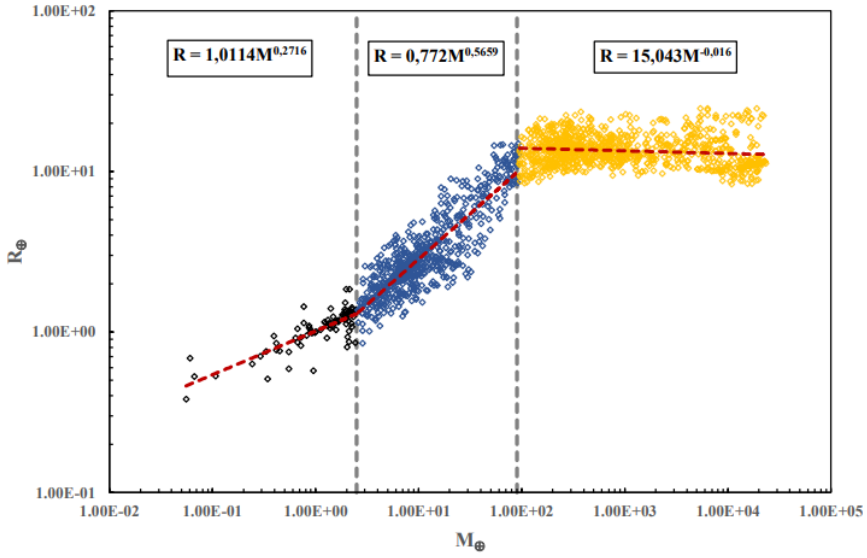


Figure 7: The M–R plot shown in Figure 6, which is divided into three sections—low-mass, intermediate-mass, and high-mass exoplanets—based on the breakpoints, and which also includes the equations describing the mass-radius relationships of these exoplanets.

These derived relationships can be used to make an approximate estimate of one parameter, such as radius or mass, for an exoplanet for which only one of these values is currently known. In addition, we can make predictions about what type of exoplanet it might be, even if we do not yet know its type, and speculate on what kinds of matter or elements it might contain.

5 Discussion and Conclusion

Mass and radius are the two most important physical parameters used to characterize celestial bodies; they provide crucial information about their formation, structure, and evolution. These two parameters are also used for the same purposes in the characterization of exoplanets. In particular, the relationship between these two parameters (M–R) provides insight into their density distribution, thereby offering information about their composition (such as rock, gas, or ice) and the type of material they are made of. Consequently, this relationship is utilized in the classification of exoplanets. As the amount of exoplanet data continues to grow, more precise results can be obtained because the methods for determining their parameters have improved. Therefore, it is very important to update studies related to such parameter distributions and relationships. In this study, the M–R relationship was examined using current data, and three

distinct M–R relationships were obtained for three different regions of the M–R distribution. As can be seen from the available literature, similar studies have been conducted previously using different methods. In this study, however, the M–R relationships were derived using a very simple and practical method that leverages the calculation and analysis capabilities of Microsoft Excel. The results obtained are very close to those derived by Müller et al. (2024) which were obtained using more complex methods.

While previous studies obtained the relation $R \propto M^{0.27}$ for low-mass planets, our study yielded $R \propto M^{0.2716}$. For intermediate-mass planets, previous studies reported $R \propto M^{0.67}$, whereas our analysis yielded $R \propto M^{0.5659}$. For high-mass planets, previous studies found $R \propto M^{-0.06}$, compared with $R \propto M^{-0.016}$ obtained in our study. These minor differences may arise from variations in the number of data points used, the selection of breakpoints in the mass–radius distribution, and the analytical methods employed.

6 Recommendations

Every day, newly confirmed exoplanets are added to the database used in the methodology part of the project, where parameters such as mass and radius of exoplanets are presented, and as of September 30, 2024, there are 7331 confirmed exoplanets in 5063 planetary systems, and 1028 of these planetary systems have more than one planet. In any project similar to this research project on the relationship between the mass and radius of exoplanets, this data continuity is crucial. As science evolves, the sheer volume of data that will be used in the project will change the graphs and analyses generated.

Today, there are other databases other than the one we use, www.exoplanet.eu, where the parameters of exoplanets are presented. Although they are presented in different databases, the exoplanets and their parameters are similar. Nevertheless, such studies can be made more comprehensive by using other databases. The features of the Microsoft Excel program were used for the analyses. For more precise analyses, comparisons can be made using problem specific programs or other available calculation and analysis methods.

Such studies can also be done with the use of artificial intelligence, which is becoming more and more widespread today. With artificial intelligence, data can be simulated by scaling with confidence, and more accurate predictions can be made by comparing with previous data. Since there is a large data set on exoplanets, artificial intelligence can analyze this data faster.

More information about the area of research can be obtained in the future and the continuity of research and projects on exoplanets can be ensured. Each new approach developed will allow us to expand our knowledge and reach solid scientific conclusions.

7 Acknowledgments

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